

Internship Report – Day 12

Date: 17/02/2026
Department: Cloud & Networking
Intern Name: Syed Fuzail Abdullah

Topic Covered: Networking Fundamentals, CIDR Calculation, Subnetting, and VNet Design

On Day 12 of my internship, I learned fundamental networking concepts including IP addressing, CIDR notation, subnetting, host bit calculation, and decimal-to-binary conversion. These concepts are essential for designing secure and scalable cloud networks in platforms such as AWS and Microsoft Azure.

1. Introduction to Networking

Networking enables communication between devices using unique identifiers known as **IP addresses**. Each IP address consists of:

- **Network Portion** – Identifies the network
- **Host Portion** – Identifies devices within the network

Example:

192.168.1.10/24

- /24 represents CIDR notation
 - First 24 bits → Network
 - Remaining 8 bits → Host
-

2. IP Address Classes

IP addresses are categorized into different classes:

Classes	Range	Default Subnet Mask	Usage
A	1.0.0.0 to 127.255.255.255	255.0.0.0	Large networks, e.g., ISPs
B	128.0.0.0 to 191.255.255.255	255.255.0.0	Medium-sized networks
C	192.0.0.0 to 223.255.255.255	255.255.255.0	Small networks, e.g., home routers
D	224.0.0.0 to 239.255.255.255	N/A	Multicast addresses
E	240.0.0.0 to 255.255.255.255	N/A	Reserved for future use

A	0 – 126	255.0.0.0	Large networks
B	128 – 191	255.255.0.0	Medium networks
C	192 – 223	255.255.255.0	Small networks
D	224 – 239	N/A	Multicast
E	240 – 255	N/A	Experimental

3. CIDR (Classless Inter-Domain Routing)

CIDR notation defines how many bits belong to the network.

Example:

192.168.1.0/26

Formulas Learned

- **Host Bits = 32 – CIDR**
- **Total IP Addresses = 2^n**
- **Usable Hosts = $2^n - 2$**

(where n = number of host bits)

Example:

/24 → Host bits = 8

$2^8 = 256$ total IPs

$256 - 2 = 254$ usable hosts

4. Subnetting

Subnetting divides a large network into smaller logical networks for better management and security.

Example:

192.168.1.0/24 divided into /26

Creates 4 subnets:

- 192.168.1.0/26
- 192.168.1.64/26
- 192.168.1.128/26
- 192.168.1.192/26

Each subnet:

$2^6 = 64$ total IPs

$64 - 2 = 62$ usable hosts

5. Decimal to Binary Conversion

Each IP address contains four octets (8 bits each).

Example:

192.168.1.1

Binary form:

Decimal	Binary
192	11000000
168	1010100 0
1	0000000 1
1	0000000 1

Final Binary Representation:

11000000.10101000.00000001.00000001

6. Practical Task – Virtual Network (VNet) Design

As part of the practical exercise, we were assigned a task to design and plan a **Virtual Network (VNet)** with an optimal IP address range and create multiple subnets based on required host capacity.

Requirement Given

Create a VNet containing three subnets:

Subnet Name	Required IPs
Test	30 IP addresses
Dev	30 IP addresses
Cloud	60 IP addresses

Subnet Calculation

To determine subnet sizes, CIDR calculations were performed:

✓ Subnet for 30 Hosts

Formula:

$$2^n - 2 \geq 30$$

$$2^5 = 32 \text{ addresses}$$

$$32 - 2 = 30 \text{ usable hosts}$$

Required subnet:

/27 (32 IP addresses)

So:

- **Test subnet** → /27
 - **Dev subnet** → /27
-

✓ Subnet for 60 Hosts

$$2^6 = 64 \text{ addresses}$$

$64 - 2 = 62$ usable hosts

Required subnet:

/26 (64 IP addresses)

So:

- **Cloud subnet** → /26
-

Example VNet Address Plan

Chosen VNet range:

10.0.0.0/24

Subnet allocation:

Subnet	CIDR	IP Range
Cloud	10.0.0.0/26	10.0.0.0 – 10.0.0.63
Test	10.0.0.64/27	10.0.0.64 – 10.0.0.95
Dev	10.0.0.96/27	10.0.0.96 – 10.0.0.127

This planning ensured:

- Efficient IP utilization
 - Proper network segmentation
 - Future scalability
-

7. Key Concepts Learned

- CIDR calculation and prefix understanding
- Host bits and usable host calculation
- Subnetting techniques
- Binary conversion of IP addresses
- Designing VNets using real-world requirements

- IP planning for cloud environments
-

Conclusion

Day 12 of the internship strengthened my understanding of networking fundamentals and practical subnet design. I learned how to calculate CIDR ranges, determine host requirements, and design a Virtual Network with multiple subnets based on real-world scenarios. These concepts are essential for cloud architecture, secure infrastructure deployment, and network segmentation in AWS and Azure environments.

Here is your **updated Day 12 Internship Report** with the additional practical task included professionally:

Internship Report – Day 12

Date: (Mention Date)

Department: Cloud & Networking

Intern Name: Syed Abdullah

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Required subnet:

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So:

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 - **Dev subnet** → /27
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Required subnet:

/26 (64 IP addresses)

So:

- **Cloud subnet** → /26
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Example VNet Address Plan

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Test	10.0.0.64/27	10.0.0.64 – 10.0.0.95

Dev	10.0.0.96/27	10.0.0.96 – 10.0.0.127
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If you want, I can next make a **more architect-level version** (which impresses internship evaluators more) or format it exactly like your previous Day 9 report style.

Microsoft Azure portal interface showing the "Create virtual network" wizard. The "IP addresses" tab is active, displaying a table of subnets:

Subnets	IP address range	Size	NAT gateway
test	10.0.0.0 - 10.0.0.31	/27 (32 addresses)	-
prod	10.0.0.64 - 10.0.0.95	/27 (32 addresses)	-
cloud	10.0.0.128 - 10.0.0.191	/26 (64 addresses)	-

The address space is 10.0.0.0/24 (256 addresses). Navigation buttons: Previous, Next, Review + create.

Microsoft Azure portal interface showing the "Review + create" step of the "Create virtual network" wizard. A green banner indicates "Validation passed".

Basics

Subscription	Vanguard Class Training
Resource Group	syedfuzail2009-rg
Name	fuzail
Region	East US

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/24 (256 addresses)
Subnet	test (10.0.0.0/27) (32 addresses)
Subnet	prod (10.0.0.64/27) (32 addresses)
Subnet	cloud (10.0.0.128/26) (64 addresses)

Navigation buttons: Previous, Next, Create. A link for "Download a template for automation" is also present.